

# Yuchen Zhou

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## EDUCATION

### Carnegie Mellon University

Master of Science in Information Networking | GPA: 3.86/4.0

Pittsburgh, PA  
Aug. 2025 - Dec. 2026

### Nankai University

Bachelor of Engineering in Information Security | GPA: 3.71/4.0

Tianjin, China  
Sep. 2021 - Jul. 2025

## SKILLS

**Languages:** C++, Python, Java, Go, C, Rust, SQL (PostgreSQL, MySQL), NoSQL (Neo4J, DynamoDB), Shell, HCL

**System & Backend:** Git, OOP, System Design, CI/CD, Linux, Redis, Kafka, Hadoop, MapReduce, Docker, Kubernetes, Terraform, FastAPI, Database Internals, Distributed Systems, Microservices, Concurrency Control, Query Optimization

**Cloud & AI Infra:** AWS (Bedrock, S3, ECS Fargate, DynamoDB, etc.), GCP, PyTorch, Claude Code, Codex, MCP

## PROFESSIONAL EXPERIENCE

### ENFOS Inc.

Software Development Engineering Intern

Durham, NC  
May. 2026 – Present

- Designed and built a privacy-aware, multi-tenant data intelligent platform across PostgreSQL, ClickHouse, and Neo4J, processing production activity for 13K+ users across 35 tenants and modeling ~20M graph relationships.
- Developed incremental, idempotent data ingestion pipelines and a persona engine that precomputed 140+ behavior features per user, enabling analytics with low latency by eliminating full-table scans over 50M+ audit events.
- Optimized DynamoDB access-pattern and secondary-index for contextual agentic memory using ordered and sparse GSIs, achieving a 95% decrease in evaluated recall rows and 44% in conversation-list latency.

### Chinasoft International

Backend Software Development Intern

Beijing, China  
May 2023 - Jul. 2023

- Drove backend Hive/HDFS-based MapReduce workflow over 4M+ records, boosting query performance by 32%.
- Delivered Spring Boot backend services and REST APIs for interactive data analytics and virtualization while coordinating closely with frontend engineers and non-technical designers to ship product features.
- Engineered MySQL data ingestion and streaming systems using Sqoop, Kafka, and ZooKeeper, ensuring reliable backend data flow and reducing end-to-end processing latency by 27%.

## ACADEMIC PROJECTS

### Relational Database Systems – BusTub DBMS ([GitHub](#))

Carnegie Mellon University

- Implemented core DBMS internal components in C++, including a disk-oriented buffer pool manager, B+ tree indexing, transaction-level and batch-based query execution engine, and optimization rules for concurrency control.
- Extended BusTub's rule-based optimizer with 5+ plan rewrite rules, including filter-scan merging, SeqScan and OrderBy to IndexScan selection, NestedLoopJoin to HashJoin, Sort-Limit-to-TopN optimization.
- Built performance-critical operators including external merge sort, window functions, and GRACE Hash Join with disk-oriented intermediate pages, improving benchmark performance by 56% and ranking in the top 5% leaderboard.

### Distributed Bitcoin Miner

Carnegie Mellon University

- Developed a custom reliable transport layer (LSP) protocol with Go, ensuring robust (sliding window protocol and retransmission), and in-order UDP communication without data races with >99% delivery accuracy.
- Built a distributed and parallel system with dynamic mining task reassignment atop LSP, decreasing re-computation overhead by 43% to increase system throughput and improving fault tolerance under node crashes.

### Distributed Scholarly Search Engine

Carnegie Mellon University

- Implemented a distributed scholarly search engine on a 3-node GCP cluster microservices using Docker, Kafka, MapReduce, and Terraform, processing 10K+ IEEE abstracts parsed from 1500+ pages of Google Scholar results.
- Improved query execution efficiency versus provided baseline solution by 37.2% by conditionally skipping MapReduce jobs for search-only requests, while supporting asynchronous multi-user query workflows on GCP.